Synchronous Sympathy at the Symphony: Conductor and Audience Accord

Teresa Marrin Nakra & Brett F. BuSha
The College of New Jersey

This study compared the emotional intensities of a conductor and audience during a concert of the Boston Symphony Orchestra. The affective state of the conductor was estimated unobtrusively with a wearable electrocardiogram, and audience participants self-reported their affective states with manual slider boxes. Results indicated that: 1) the conductor's heart rate variations were temporally aligned with structural patterns in the musical scores; and 2) these variations strongly correlated with the average emotional intensity measurement of the audience. Four consecutive musical selections yielded significant positive correlations ($p < .001$; $r = .86, .84, .48, .61$), demonstrating a temporally related emotional intensity shared between the conductor and audience during the performance. Although a causal relationship was not determined, some evidence supported induction as the mechanism of emotional communication. These results suggest possible methods for better understanding the affective experiences of conductors, the reactions of listeners to various stimuli, and the interactions between audiences and musicians during concerts.

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People commonly report experiencing affective responses to music and cite these among their primary reasons for engaging in musical activities (Eerola & Vuoskoski, 2013; Juslin & Laukka, 2004). Listeners' emotional responses to music have been found to be continuous, interactive, interrelated, time varying, and complex (Becker, 2001). Musicians also report experiencing emotional responses while performing. However, little is known about the nature of musicians' emotional responses and how they might correlate with either audience responses or musical structure in a live concert setting.

In order to better understand and compare the experiences of musicians and audiences, new approaches are needed. Jaimovich, Coghlan, and Knapp (2010) addressed this issue by undertaking a pilot study that investigated physiological convergence between performers and audience members. Their preliminary results indicated significant correlations between the physiological states of performers and audiences, which they ascribed to emotional contagion from musicians to audience members. They further proposed that recent work on mirror neurons could possibly explain the mechanism of contagion.

Eerola and Vuoskoski (2013) recently analyzed 170 listening studies related to emotion, and determined that none of them were conducted during a live performance. They concluded that researchers should "venture outside the laboratory to study emotions in the contexts in which they naturally occur" and ensure that stimulus material is "ecologically valid." They also advocated combining multiple measurement approaches, including continuous self-report and biological sensors, to accurately characterize the temporal nature of music-induced emotions.

Some researchers have investigated the emotional intentions and experiences of musicians (Baraldi, De Poli, & Roda, 2006; Gabrielsson & Juslin, 1996; Juslin & Laukka, 2003); while such studies have generally used classical music stimuli, most have not measured conductors or listener responses. Wöllner and Auhagen (2008) measured the conducting expressiveness of advanced student conductors through continuous judgements of trained musicians from different visual perspectives within the orchestra. Theirs may have been the first such project to investigate relationships between movements of conductors and continuous responses by observers, although the auditory/visual stimuli were recorded, and the instruments of the orchestra simulated by two pianos.

While it can be challenging to balance the opposing demands of experimental control with the complexity and unpredictability of research in a realistic setting, it is important to undertake research on music and emotions in the context of live performances. When carefully applied, physiological and behavioral sensors

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can be used to accurately measure emotional responses in these settings. These sensors can work well for both performers and listeners, due to the immediate and reflexive nature of these responses. Most researchers agree that the intensities of emotional responses to music can be reliably measured and analyzed (Hodges, 2010; Sloboda & Lehmann, 2001), although there is some ongoing debate in the research literature about whether emotional intensity is actually simply the quantity known as arousal in the two-dimensional orthogonal (arousal-valence) "circumplex model" of emotion first proposed by Russell (1980) (Eerola & Vuoskoski, 2013; Schmidt & Trainor, 2001). For the purposes of this study, emotional intensity has been defined as the strength or force of a felt emotional response, regardless of specific concomitant arousal and valence components. High intensity emotions could be of any type (e.g., intensely happy, intensely sad, etc.), whereas low emotional intensity could be characterized by terms such as apathy or boredom. (Participants in this study were given instructions that reflected this definition, as described in the Method section below.)

Heart rate (HR) is a physiological measure that can be modulated by emotional intensity and induced by music (Nyklicek, Thayer, & Van Doornen, 1996) or imagery (Schwartz, Weinberger, & Singer, 1981; Sinha, Lovallo, & Parsons, 1992). Experimental data suggest that emotional intensity elevates HR similarly across targeted evoked emotional responses, when compared to a control condition. HR can be measured unobtrusively by recording electrocardiogram signals with surface electrodes; this noninvasive method does not require a subject's directed input. Alternatively, emotional intensity can be measured with behavioral self-assessments, using self-report mechanisms that require participants to assess and register their responses in real-time. Standard measuring devices for self-report include manual interfaces such as dials, slider knobs, and pressure sensors (Poh, Swenson, & Picard, 2010).

The research literature in music perception indicates much debate about the mechanisms by which affective responses to music are synchronously shared by listeners and musicians. Some theorize that affective responses are *induced* or contagiously communicated in listeners (Goleman, 1995), sometimes called the *emotivist* position. Studies supporting induction indicate that listening to music elicits clear physiological responses that correlate to emotions (Baumgartner, Esslen, & Jäncke, 2006; Krumhansl, 1997). In contrast, the *cognitivist* position posits that affective responses to music are instead reflections of the *recognition* or *perception* of emotion in music by listeners. The relative merits of the cognitivist and emotivist positions have been thoroughly addressed in the literature (Kivy, 1989, 1990), but are not well understood in the context of live performance. For example, it has not been established whether a conductor (or other musician) must genuinely feel (self-induce) emotional responses in order to communicate or share them effectively. Some theorists and music critics have suggested that experienced professional musicians create a cognitive *proxy* for emotions, learning to express the forms or shapes of those emotions without needing to directly experience them: "a conductor's job, like an actor's, is not to experience an emotion directly but to know what it is and how to communicate it to others." (Dyer, 2006) One reason why musicians might purposefully avoid experiencing emotions while performing might be to avoid any physical effects (increased respiration rate, muscle tremors) that could negatively impact the quality of the performance. It seems that this phenomenon has not yet been investigated in music perception research.

One important unifying element in an orchestral concert is the musical score; it is the information source that conductors and musicians use to create a performance. Some musicians characterize the way in which they interpret and perform those scores as providing alternating moments of varying degrees of tension and release to vary the intensity of the music for the purpose of engaging an audience. Conductor David Epstein once wrote, "Tension means energy unresolved, and unresolved energy ultimately means forward motion. Indeed, this is the purpose of it all." (Epstein, 1998, p. 7) Some musicians call this strategic and purposeful variation of intensity *interpretation*, and relate it to structural properties in the score (White, 2013).

It has long been assumed that listeners experience complementary responses to the interpretational variations of musicians. To explain these concomitant responses in listeners, Vines and colleagues proposed that listeners store and release "affective energy" in potential or kinetic forms, depending on local patterns of musical tension (Vines, Nuzzo, & Levitin, 2005; Vines, Wanderley, Krumhansl, Nuzzo, & Levitin, 2004). Vines explored how music creates in the listener a feeling of "affective velocity" or "affective acceleration," describing in more detail the dynamic contours and trajectories that audiences experience while listening to music. Huron (2006) and Narmour (1999) have provided alternate models to explain the continuous affective experiences of listeners.

Several researchers have attempted to demonstrate the affective characteristics in musical scores by modeling the emotional intensities encoded in patterns of

notes. Some approaches have looked at how time-varying patterns of tension and resolution in a listener's experience correlate with patterns of tension and release in harmonic, melodic, and rhythmic structures. Farbood (2012) combined and extended several existing frameworks to create a quantitative, multi-parametric model that combines musical tension, harmonic tension, melodic expectation, and perceived loudness with basic factors such as pitch, time, and speed. This approach presents a potentially significant method for anticipating the listening experience with regard to tension and resolution.

If Farbood's (2012) model of musical tension is largely accurate, then it might be assumed that listeners perceive and respond to tension information continuously over the course of a piece of music or an entire concert. As was shown by McAdams, Vines, Vieillard, Smith, and Reynolds (2004), these cognitive audience responses to musical tension can be approximated over time using continuous statistical estimates of an audience's collective "emotional force" (emotional intensity) response. During several live concerts, McAdams measured audience emotional force responses that aligned closely with large-scale sectional structure in the musical scores. Participants operated slider boxes with one-dimensional continuous tracking mechanisms while listening to a new orchestral work composed specifically for that experiment. More recently, Upham has undertaken a comprehensive review of analytical techniques for understanding the temporal dynamics of slider data collected in studies of this kind (Upham, 2011; Upham & McAdams, 2009). Methods for working with continuous audience emotional intensity data are not yet conclusively established; for this reason, this study adopted a more traditional statistical approach using cross-correlations to compare and characterize temporal relationships between continuous response sequences. This technique is detailed in the Results section; strengths and weaknesses of this approach are assessed in the Discussion section below.

Compared with the substantial number of studies of tension (or affective intensity) during music listening, there are very few examples of research on musicians' affect while performing or conducting. Gerhart Harrer was perhaps the first researcher to place sensors on an orchestral conductor; he worked with Herbert von Karajan, the longtime conductor of the Berlin Philharmonic (Harrer, 1982). Harrer investigated von Karajan's HR and other physiological responses (respiration, electrodermal activity, temperature) under different conditions, including listening to and conducting a single piece — Beethoven's *Leonore* no. 3 Overture. Harrer found strong correlations between Karajan's physiological responses and structural events in the music, and noted that the strongest changes in the conductor's signals occurred during the moments when Karajan reported the highest affective intensity. Harrer also noticed that Karajan's responses were less intense while listening than while conducting. While Harrer acknowledged that a greater amount of movement and physical effort was needed for conducting than for listening, he determined that much of the difference could be attributed to the fact that HRs during listening were in reaction to stimuli, whereas HRs during conducting reflected a purposeful and premeditated expression. While Harrer did not conclusively determine whether the physical movement of conducting caused the conductor's HR to change, or whether HR was influenced by expressive musical intentions or emotions, his research nonetheless pioneered new methods for studying conductors' affective experiences.

The electrocardiogram worn by the conductor in this study was integrated into an updated version of the *Conductor's Jacket*, a wearable, lightweight, and unobtrusive sensor-infused garment (Nakra, 2000; Picard & Healey, 1997). The original *Conductor's Jacket* measured physiological signals from professional and student conductors during orchestral rehearsals and performances, demonstrating correlations between orchestral conductors' internal states and events in a musical score (Nakra, 2000). The authors of this study have jointly continued to build wearable systems that capture physiological data from musicians during rehearsals and performances (Rahimi, Mastro, Nakra, & BuSha, 2012). The current study was designed to extend prior work by measuring the degree to which the conductor's internal experience and the audience members' responses aligned.[1]

There are many individual contributors to a professional orchestra performance, potentially complicating the transmission of affective information from conductor to audience. In thinking about how to properly model this situation in order to effectively measure it, the study authors found a useful analogy in the classic information theory approach proposed by Shannon

[1.] A followup study was conducted two months after this study by a team led by Daniel Levitin and Stephen McAdams at McGill University. Sixty members of a remote audience experienced the same orchestral performance via high definition digital video and audio rebroadcast in a concert hall, while operating the same continuous tracking slider boxes. The purpose of the followup study was to compare the audience responses between the live and recorded settings. The results indicated similar features across both audience responses but greater average intensities in the live audience as compared with the video rebroadcast (Upham & McAdams, 2009; Upham, 2011).

(1948). From Shannon's perspective, the communication of affective information between a conductor, orchestra, and audience could be understood to take place in successive, nearly synchronous stages, with the musical score serving as the information source (providing a contextual and structural reference, a kind of *ground truth* against which the received message could be compared). In this model, the role of the conductor is to express the interpreted score information (including emotional intensity) while coordinating and shaping the musicians' actions. The musicians receive information from the conductor while reading their own individual scores and producing the sounds. Members of the audience receive auditory and visual stimuli and generate their own internal responses.

Preparations for orchestral concerts can sometimes begin long in advance, with a process of score study and interpretation by the conductor. Much closer to the concert date, the conductor and musicians rehearse the interpreted score, bringing the notes to life and shaping them through physical and cognitive effort. According to Shannon, communication stages of conductor, orchestra, and audience could each react to received information, add new information to the message, and pass it along to the others. There could be delays, attenuations, and noise, as well as secondary interactions between stages. This analogy with information theory provided a useful framework with which to model and plan the current study.

Motivation

The intensities of listener responses to recorded music have been well documented in music perception research. However, with some exceptions, little is understood about how emotional responses are generated in a concert setting or how musicians' and audiences' responses might differ and influence each other. Since music is often experienced in live concerts, such interactions do occur and are worthy of study.

This study investigated the temporal correlations of emotional intensity responses in both performers and audience members in a classical music concert. The aim was to quantitatively determine the emotional experiences of performers and listeners, investigate whether there was a measurable relationship to the score, and establish whether the experiences of performers and listeners were temporally correlated. The approach was to induce a measurable change in emotional intensity from the audience, as self-reported, in response to (or in synchrony with) a correlate of emotional intensity, measured as a change in HR in the conductor. It was predicted that commonly accepted measures of affective response would yield measurable correlations between conductor and audience. Since both physiological responses and self-report measures can be used to determine emotional intensity, those quantities were measured and compared across conductor and audience. It was hypothesized that conductor and audience would experience emotional intensity trajectories that could be correlated and linked to features in the score.

Method

PARTICIPANTS

The data used in this study were collected on April 8, 2006, during a public concert performance of the Boston Symphony Orchestra (BSO). The conductor and thirty-two audience participants provided continuous measurements of their emotional intensity during the performance, and these data were compared afterwards. The audience sample was somewhat diverse and did not have high levels of music training; the average age was 32, with the youngest aged six and the eldest aged 71 (ten participants were aged 12 or under). Of 30 who reported their gender, 18 were female and 12 male. The average number of years of music training was 3.7 (with nine respondents reporting no training), and the average amount of daily music listening was nearly 1.5 hours, across many different genres.

PROCEDURE

Due to significant cognitive and physical performance demands, the conductor's emotional intensity was measured noninvasively with wearable electrocardiogram monitoring. The conductor wore sensors that recorded the electrical activity of the heart from electrodes on the skin surface (ECG). With the movements required to conduct a full orchestra, it was essential to ensure that electrical sensor leads were securely attached and not subject to stray motion artifacts. The ECG sensor used in the BSO concert was integrated into an updated version of the *Conductor's Jacket* (Nakra, 2000).

The audience participants' emotional intensities were obtained behaviorally from each individual via self-reported assessment. Audience participants operated manual input slider boxes with continuous tracking, loaned from the Music Perception and Cognition research team at McGill University (Upham & McAdams, 2009). The slider boxes were set up in the front left section of the audience nearest the stage, to minimize cable lengths to the computer backstage. The method established by McAdams et al. (2004) was used; participants were asked to provide their

felt emotions rather than their subjective perceptions of the *expressed emotions* of the musicians. Participants continuously adjusted the slider to report their ongoing, real-time responses. The following written instructions were given to the audience members who volunteered to participate in the slider box study:

> "You'll be using a 'slider box' to record your emotional response to the music performed by the BSO. The slider boxes are very easy to use. During the performance, indicate the degree of your 'emotional intensity' by moving the slider between the 'Low' and 'High' positions. Basically, if you feel strongly emotional—happy, sad, tense, *whatever* the emotion—move the slider toward the 'High' position. If you feel less emotional (apathetic, bored, unmoved), move the slider toward the 'Low' position. Moderately emotional? Move the slider toward the center. Remember, you're recording the intensity of your *own* emotional experience, not necessarily the intensity, volume, or emotional content of the music itself. We want to record your actual, personal reaction to the music."

Audience participants read instructions and interacted with trained staff members before the concert. They also received a concert program that explained the concepts and methods of the experiment, including this: "the information gathered today will be used to understand how musicians create emotional intensity through their movements and gestures, and how an audience receives and reacts to that emotional communication."

Research methodologies were reviewed and approved prior to the experiment by the BSO Musicians' Committee and the Institutional Review Boards at McGill University and The College of New Jersey. An extensive custom sensor data acquisition system was built for this project, designed to accommodate a large number of moving and stationary participants, located in both the audience seating area and the stage. All data were recorded onto to a single computer backstage. Three pilot studies were undertaken to test the sensor hardware and train technical staff. The Boston Symphony Orchestra did not hold a rehearsal to test the sensor hardware in the hall. Informed consent was obtained for all participants, who were instructed in the proper placement and use of their sensing devices. Twelve trained technical staff and ten university students assisted with participant releases, instructions, sensors, and data collection. To document the event, three video recordings captured different views of the conductor, orchestra, and audience, and a professional quality audio recording was produced. A musical analysis was applied afterwards to compare the emotional intensity features of the conductor and audience with actual musical features in the score, as determined by a conventional harmonic and structural analysis.

MATERIALS

The Boston Symphony Orchestra (BSO) performed four consecutive works by W.A. Mozart during the first half of the concert, totalling 18 minutes of music:

- Overture to *The Marriage of Figaro*, K.492
- *Symphony No. 1* in E-flat (movement 3 *Finale*), K.16
- Concerto for Clarinet in A (movement 2 *Adagio*), K.622
- Symphony No. 41 in C (*Jupiter*, movement 4 *Finale*), K.551

DATA PREPARATION

The data were successfully recorded from the conductor and 32 audience participants during the concert; a backstage computer recorded the 33 independent channels into individual data files for each of the four musical pieces. The conductor's ECG sequences were first converted to HR information offline by measuring the time interval between successive R-waves (the largest positive waveform in the ECG related to a single heartbeat). There was a small amount of noise in the ECG signal due to a bandwidth mismatch between an isolation transformer (20-1,800Hz) and the ECG sensor (2-30 Hz). The incoming signal approximated an impulse input to the transformer, which caused a ringing effect in the recorded signal; however, it was determined that this noise would not impact the HR calculation.

Next, pre-processing was required in order to correct a slight error introduced into the data related to sampling rate variability. Each sensor was sampled over the duration of the concert at approximately 877 Hz, with average sampling rates across the four pieces drifting between 855 Hz and 898 Hz. This variability in the sampling frequency was attributed to the Windows clock timestamp used in the data acquisition software. The number of samples collected per second varied, despite a clock timestamp with a time resolution of one second. Table 1, below, summarizes the average sampling rates and standard deviations for each of the musical pieces that were studied.

This variability in the average sampling frequency complicated the measurement of the HR intervals. A new method was developed to compensate for the drifting sampling rate by resampling the ECG data on a second-to-second basis. Taking the actual number of

TABLE 1. *Sampling Rate Averages and Standard Deviations for Each of the Musical Selections.*

piece	Figaro	Sym1	Clarinet	Jupiter
avg rate (Hz)	891.1	893.1	855.5	898.3
std dev	66.7	41.2	88.0	94.1

samples within each sequential second, a constant sampling rate was assumed within that second. The reciprocal of each number of samples per second was taken, to calculate approximate time steps on a second-per-second basis. This step corrected for the number of samples during each sequential second, and these values were later used to determine the bins for time-aligning the audience slider data. (This method was determined to be an effective solution, because the minimum sampling rate was well more than an order of magnitude larger than the frequency of the HR events being measured.)

Once the ECG data were properly time aligned, the HR sequence timing normalization process commenced. Individual heartbeats were detected by a locally designed MATLAB algorithm (BuSha, 2010). A single characteristic R-wave was manually selected, and then the program automatically located all R-waves within each ECG file. Prior to the temporal normalization process, improper or excluded detections were corrected. After all the peaks were identified, the time points at each peak were determined. An algorithm computed the time interval between successive peaks and assigned that interval to the second peak. The resulting sequence contained interbeat distances indexed by beat number. In this representation of the data, HR was a function of variable beat-distances. Therefore, time was not constant. To index the x-axis values by time instead of beat number (and thereby prepare the sequence for comparison with the audience data), all prior heartbeat values were summed (n + n-1, n-2, etc.) for each successive value. This yielded HR over time, indexed at the correct x-axis values. The HR was then converted to units of beats per minute (BPM) by computing 60/interbeat interval. This process produced a raw (unfiltered) HR sequence for each piece. The conductor's raw HR for the first musical selection is shown in Figure 1:

As expected, the conductor's unfiltered HR contained a high amount of variability from second to second. Some of this variation was inherent to the method that was applied; instantaneous interpulse intervals determined the rate, which itself was not instantaneous, but rather a time-varying measure. The HR sequences were therefore smoothed to remove instantaneous noise and reveal larger trends, using a 5-point moving average filter. In order to avoid phase lag induction, the MATLAB *filtfilt* function was selected. (It performs zero-*phase* filtering by processing the input data in both the forward and reverse directions.) These steps obtained a clearer estimate of trends that averaged out short-time duration anomalies and attenuated some of the extreme variation demonstrated in Figure 1. The last two points of the filtered data for the *Marriage of Figaro* and *Symphony No. 1* sequences were removed in order to exclude boundary-related mathematical artifacts related to the filtering process. Figure 2 shows the conductor's HR during the first musical selection, obtained using the methods described above:

The conductor's HR sequences were then compared with the audience data for each piece. As described in Upham and McAdams (2009), the individual audience participants' continuous slider box data were quite

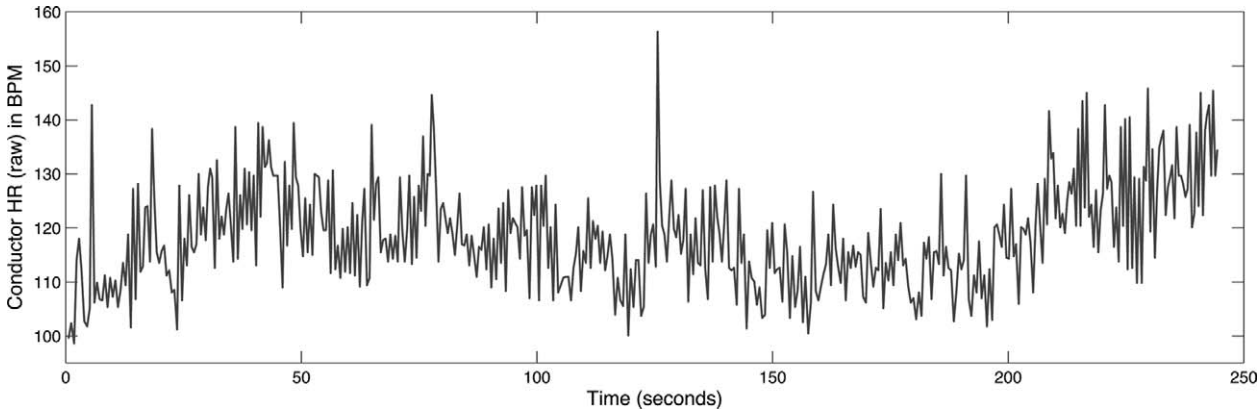

FIGURE 1. Conductor's unfiltered HR during *Marriage of Figaro.*

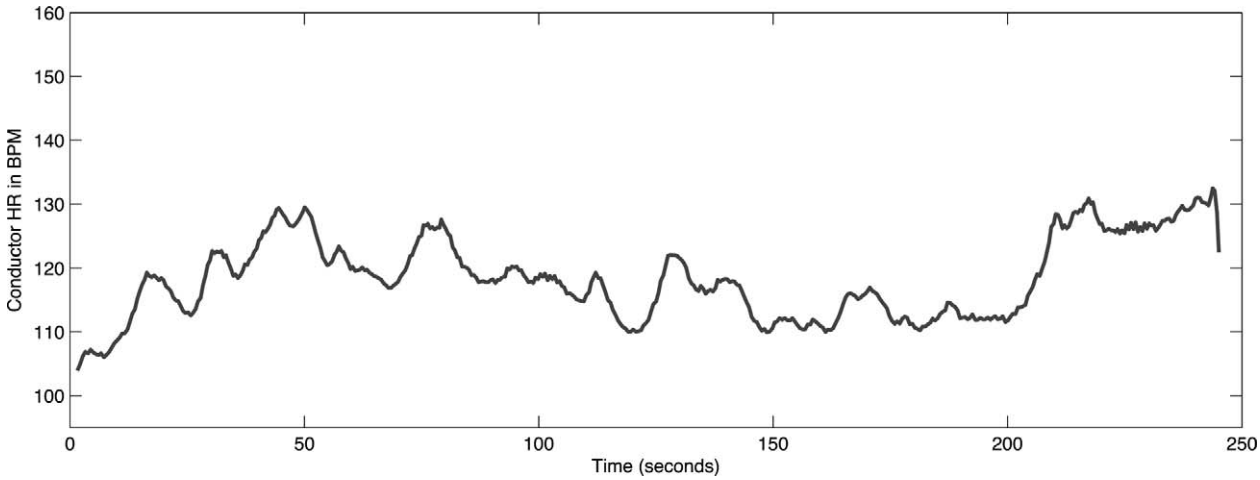

FIGURE 2. Smoothed conductor HR from *Marriage of Figaro.*

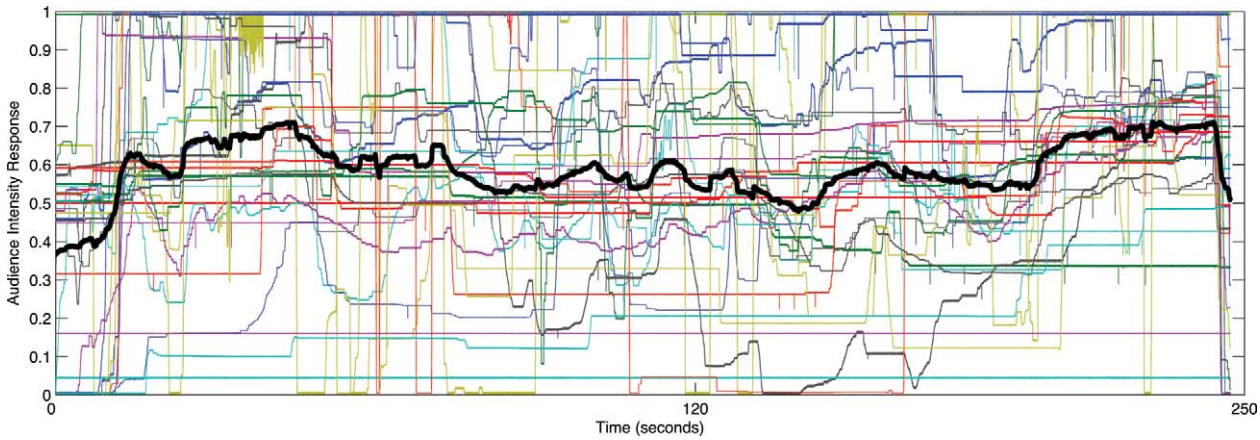

FIGURE 3. Individual audience responses (color), and their average value (black) during the performance of Mozart's *Marriage of Figaro* Overture. (See color version of this figure in the print color plate or online at ucpress.edu.)

variable. One standard approach for summarizing these data is to compute the average rating as a function of time (Schubert, 2004); this method was selected. The slider data were first pre-processed to align with the time-corrected HR data. Each slider value was associated with its corollary HR interval. Then, the average was taken across all 32 values. Finally, the same 5-point moving average filter that was used for the HR was applied to smooth the slider signal. The thirty-two audience slider box responses are presented in color in Figure 3, below, with the average value (bold) in black:

Finally, the conductor's HR sequences were compared and jointly plotted with the averaged audience response data. First, the average slider values were plotted against their own x-axis in order to preserve a constant time-scale; the audience data were plotted at regular time intervals, whereas the conductor HR was plotted against a variable interbeat distance. Both series were normalized to match their vertical scale.

Results

Figures 4-7 present the resulting data comparisons for the four pieces studied, plotted in two points per second. The blue lines denote the conductor's HR; the red lines represent the average slider value of the 32 audience participants:

The next step was to quantify any similarity between the conductor's heart rate and the average slider box data of the audience. Several analytical techniques to compare continuous temporal data sets associated with musical performance were assessed, from cross-correlation

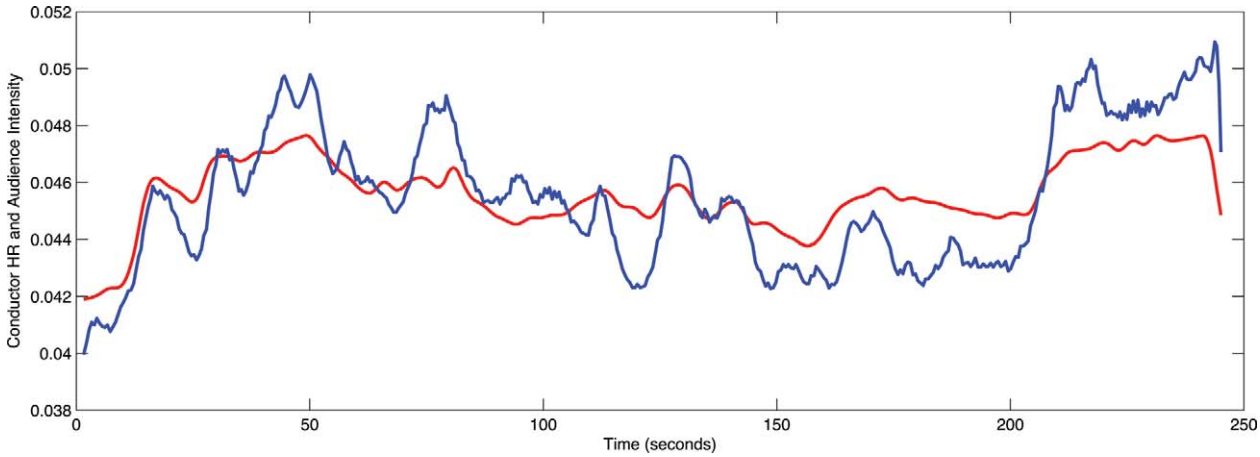

FIGURE 4. Conductor's HR (blue) and audience response (red), normalized, during the performance of Mozart's *Marriage of Figaro* Overture. (See color version of this figure in the print color plate or online at ucpress.edu.)

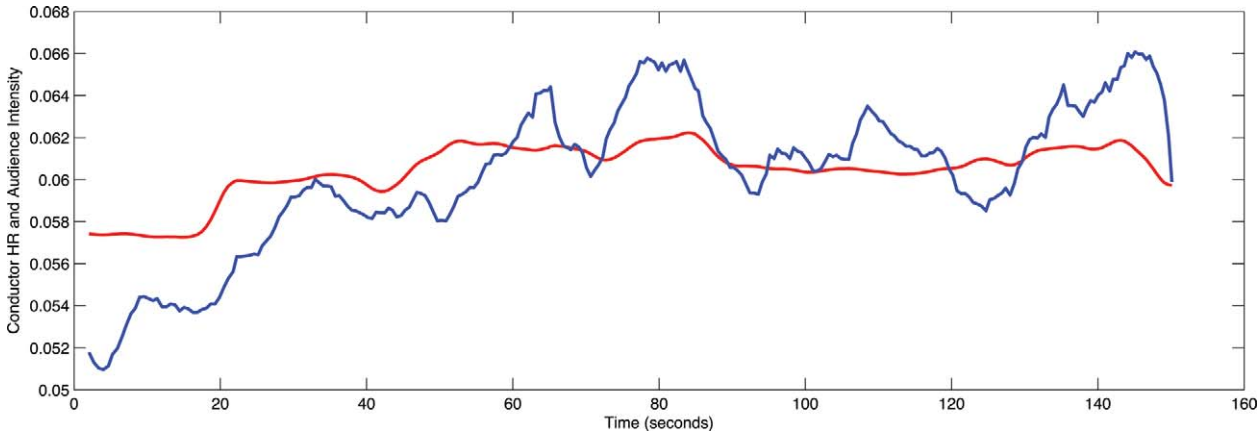

FIGURE 5. Conductor's HR (blue) and audience response (red), normalized, during the performance of Mozart's *Symphony no. 1.* (See color version of this figure in the print color plate or online at ucpress.edu.)

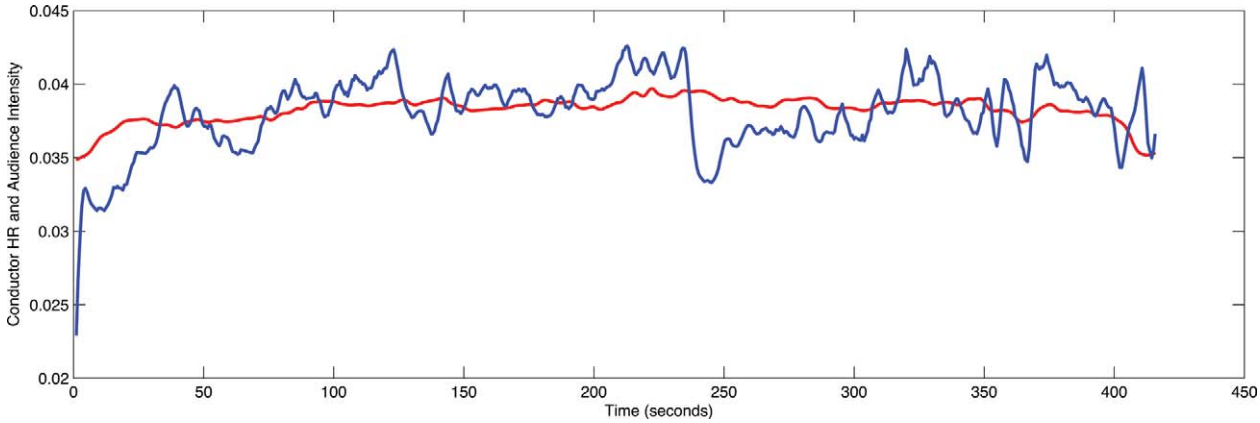

FIGURE 6. Conductor's HR (blue) and audience response (red), normalized, during the performance of Mozart's *Clarinet Concerto*. (See color version of this figure in the print color plate or online at ucpress.edu.)

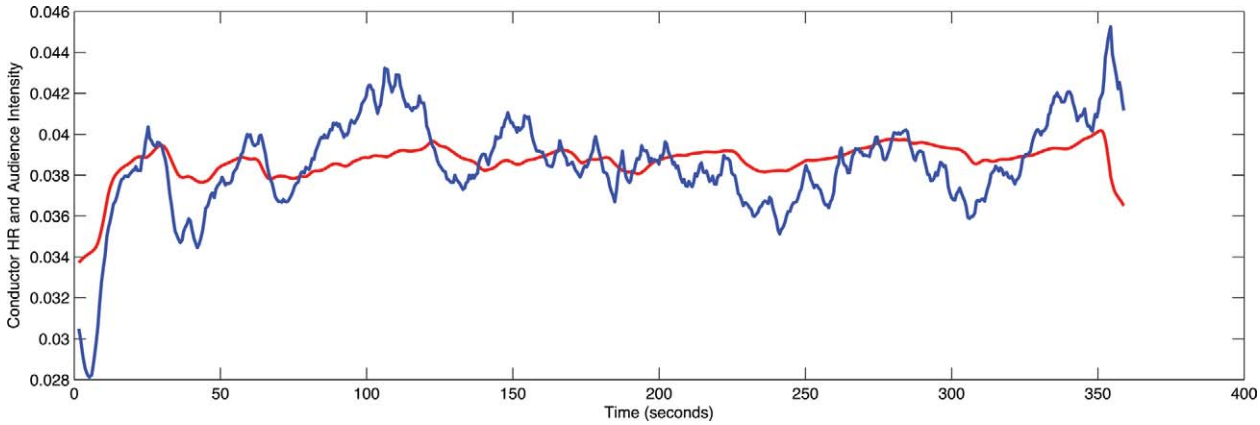

FIGURE 7. Conductor's HR (blue) and audience response (red), normalized, during the performance of Mozart's *Jupiter Symphony.* (See color version of this figure in the print color plate or online at ucpress.edu.)

(Caramiaux, Bevilacqua, & Schnell, 2010; Goebl & Palmer, 2009) to the recently proposed use of time series analysis (Bailes & Dean, 2012; Dean & Bailes, 2010, 2013). In order to minimize data transformation prior to analysis, the authors opted to use cross-correlation to quantify the similarity between the data sequences. The correlation coefficients and p values (*corrcoef* function, MATLAB) that were obtained for the four pieces are displayed in Table 2:

The following technique was then applied to verify these results: for each set of averaged slider data, ten surrogate data sets were produced by randomly shuffling the original slider data. A new correlation value and associated p value was calculated for each surrogate data set relative to the associated conductor's HR data. The surrogate data sets were constructed to determine whether the correlations between the slider and HR data from each trial were significantly different from that of the random slider inputs and the conductor's HR data. The randomized slider data sets returned average r values (for the ten surrogate data sets) of −.09, −.05, .007, and −.006 respectively, for the four concert pieces (*Marriage of Figaro*, *Symphony no. 1*, *Clarinet Concerto*, and *Jupiter Symphony*).

Overall, the results of this study indicate that the conductor's HR was highly correlated with the average audience intensity measurement over the four musical selections in the concert. The highest degree of correlation occurred during the first piece, $r(30) = .86$, $p < .001$. The first four selections in the concert were studied; all four had significant positive correlations, the average of which was $r(30) = .70$, $p < .001$. Overall, this result indicates nearly 70% correlation across 18 minutes of music, with the p values indicating a high degree of statistical significance. Features in the conductor and audience responses appeared to be temporally aligned.

TABLE 2. *Correlation Coefficients and p values for Each of the Musical Selections.*

Piece	Figaro	Sym1	Clarinet	Jupiter
Correlation coefficient	.86	.84	.48	.61
p value	< .001	< .001	< .001	< .001

The results of this research also indicate that variations in the conductor's and audience participants' emotional intensities were temporally aligned with structural features in the musical scores. A formal analysis of the *Marriage of Figaro* score was performed using standard music theory techniques such as identifying themes and keys. The *Marriage of Figaro* Overture has a slightly unusual structure; although it is often described as conforming to sonata or sonatina form, it is more correctly described as a double sequence of four distinct melodies, knitted together with transitional material. Mozart scholars agree that the Overture was written after the opera was finished, one day prior to the premiere. Unlike other opera overtures by Mozart, it does not contain any of the material from the opera, but instead brilliantly and inventively sets a bristling pace (Presto) and mood for the beginning of the drama.

Figure 8 identifies the structure of the *Marriage of Figaro* Overture. The thematic material has been grouped into four sections (identified as A, B, C, and D), with the transitional materials labeled as T1, T2, and T3. The sections are indicated by color labels, aligned with the conductor's HR signal. Beneath each colored thematic label, the primary key of the section is listed.

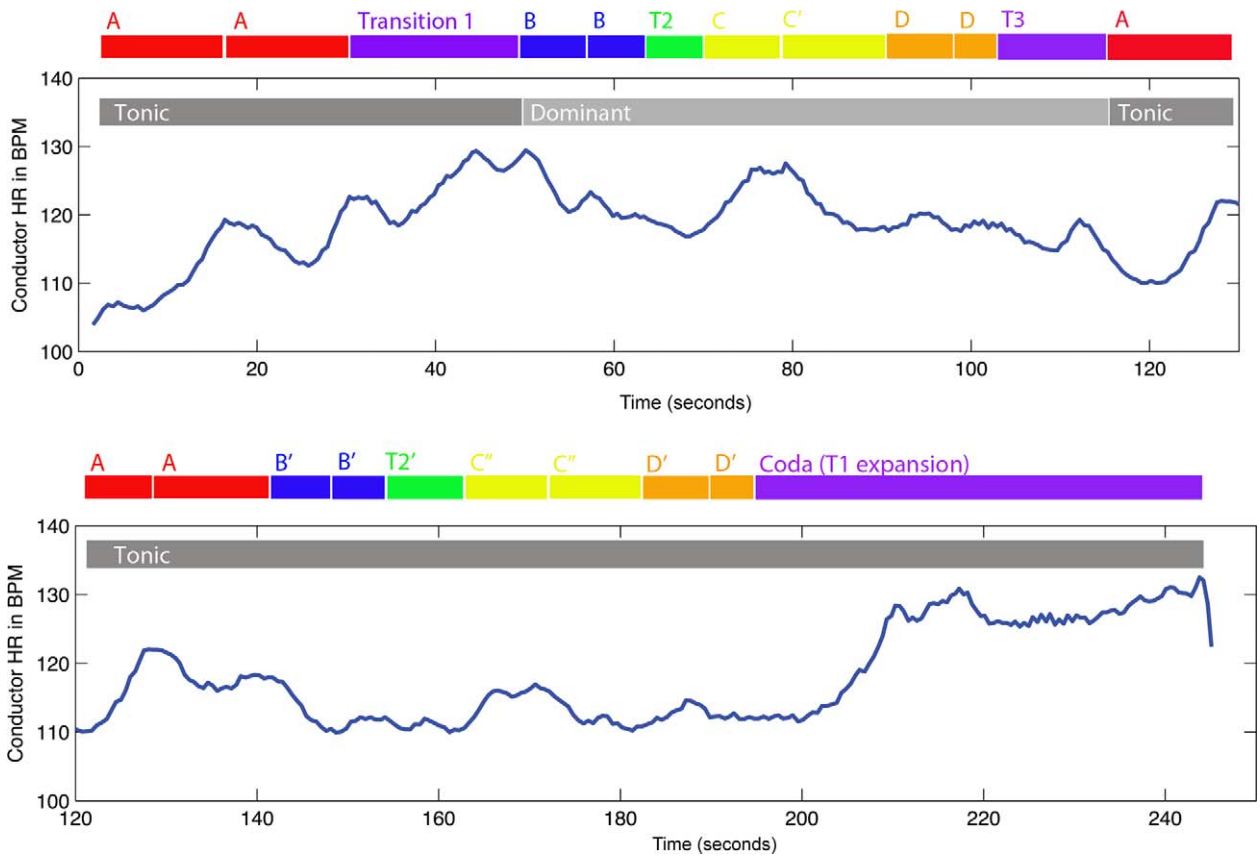

FIGURE 8. Conductor HR aligned with musical sections in *Marriage of Figaro*. (See color version of this figure in the print color plate or online at ucpress.edu.)

The piece begins in the tonic key (D Major), modulates to the dominant (A Major), and returns to the tonic. The sections have been time-aligned with reference to a video recording of the performance, which can be accessed at http://www.youtube.com/watch?v=_lAbk8cljdA.

From this visual comparison, it is clear that sections in the music correspond to trends in the HR signal. For example, the A theme and T1 transitional materials are accompanied by increases in the conductor's HR, whereas the B theme is accompanied by a falling HR. The C and D thematic materials demonstrate less consistency of correlation; the first time they appear (in the dominant key of A major), they are accompanied with HR increases; the second time they appear (in the tonic key of D major), the HR is more constant.

Discussion

This study investigated whether emotional intensity was shared between conductor and audience during a live concert of the Boston Symphony Orchestra. Commonly accepted measures were used to assess the emotional intensities of the conductor and audience during the concert, and significant correlations were found between conductor HR and audience measurement. This study also revealed evidence of strong temporal alignment between the conductor's and audience's emotional intensities. These correlations in affective state between conductor and audience indicate that a high degree of emotional synchrony occurred, and provide further justification for the use of HR as an indicator of emotional intensity. Features in the conductor's emotional intensity measurements coincided with some structural characteristics in the score itself, obtained through visual inspection of thematic and harmonic elements in the score and the performance video recording. While the results of this study do not indicate dependence or causation due to the high number of variables in the social context of an orchestral concert, they nonetheless provide strong statistical support for the claim that during a concert, the audience feels and

reports an emotional intensity contour that is highly similar to that of the conductor. While individual responses varied, the behavior of the audience as a group correlated strongly with the conductor's HR.

The highest correlation between audience and conductor responses was observed for the first piece on the concert (followed by the second piece, the fourth piece, and then the third piece); this result implies that there was a strong effect of program order on audience responses. Disagreement among participants' continuous responses to individual pieces may have contributed to this effect (Upham, 2011, p. 87). It is also possible that characteristics of the all-Mozart program might have contributed to the significant and high correlations observed. The stylistic and structural elements of Mozart's compositions are familiar to contemporary classical audiences, irrespective of music training or knowledge. Therefore, while the audience participants were not highly trained in music, it is still conceivable that some of them were engaging in a predictive cognitive process that might have influenced their responses.

The temporally correlated affective state shared between the conductor and audience indicates a high degree of *interpersonal synchrony. Synchrony* refers to individuals' temporal coordination during social interactions. The analysis of this phenomenon is complex, requiring the perception and integration of multimodal communicative signals (Delaherche et al., 2012). As has been stated previously, the research literature in music perception indicates much debate about the mechanisms by which physiological and affective responses are conveyed and synchronously shared. While the degree of synchrony fluctuated somewhat between pieces and at moments within piees, the interpersonal synchrony between conductor and audience participants seems to have been high overall.

The participant instructions used in this study provide some evidence that emotional induction occurred between conductor and audience, because participants were asked to provide *felt emotions* rather than *perceived emotions* (their impressions of the expressed emotions of the musicians): "remember, you're recording the intensity of your *own* emotional experience, not necessarily the intensity, volume, or emotional content of the music itself. We want to record your actual, personal reaction to the music." To the extent that participants accurately completed this task, the "locus of emotion" (Evans & Schubert, 2008) was in the listener's direct experience. However, the accuracy of audience participants' task completion was not assessed in this study. Also, it was not determined whether the primary stimulus came from the conductor or orchestra, and to what extent visual or auditory signals contributed to the measured responses. Therefore, while some evidence from this study seems to support induction as the mechanism of emotional communication, further investigation is needed to clarify the mechanism that caused the correlation between the *felt emotion* of the audience and the intensity experienced by the conductor.

There were some experimental and methodological limitations of the current study, including the complexity of interactions between participants during the concert, and the difference in the measurement modality between audience and conductor. The decision to use a hybrid approach combining self-reports and physiological measures was made in order to avoid interfering with the conductor's range of motion or cognitive load while performing. Another study limitation involved the relative importance of visual stimuli versus auditory stimuli to the listeners' experience. Whereas other studies of listener responses to music have investigated uni and multimodal conditions (audio-only, visual-only, and audio-visual) (Broughton & Stevens, 2009; Davidson, 1993; Vines, Krumhansl, Wanderley, & Levitin, 2006), these conditions were not measured in the current study.

Another methodological limitation might be associated with the slider box self-report method used in this study (Upham & McAdams, 2009); participants were asked to provide their *felt emotions* rather than their (subjective) perceptions of the *expressed emotions* of the musicians. Whether participants can determine and consciously move a slider to quantitatively describe their felt emotions while listening in real-time is an open question. The cognitive mechanisms that enable audiences to perceive and report affective information in music in real-time are not well understood.

Further study limitations might be ascribed to factors that could have caused variability in the conductor's HR signal, in addition to emotional intensity. One factor may have been the cognitive load of conducting itself. HR has been shown to increase with cognitive load (Rowe, Sibert, & Irwin, 1998; Wilson, 2002); higher HRs could indicate sections of music that required more intense cognitive processing for the conductor. Identifying the sections that involve higher cognitive workload for the conductor could help explain changes in the HR signal.

In addition to the cognitive workload component, physical effort has a significant impact on HR. The conductor's HRs in this study varied between 100 and 140 BPM, resembling a normal HR during moderate to intense exercise. This is consistent with Harrer's HR study (Harrer, 1982); the heart works hard during conducting to pump blood through the smaller arteries in

the arms. A method for estimating the physical effort of conductors is needed in order to be able to identify and estimate the weighted (and possibly interrelated) contributions of physical effort and emotional intensity to conductors' HR signals. One approach might be to assume that effort varies with the tempo and volume of the music. Another method might be to determine the conductor's physical effort experimentally using additional sensor data. Physical effort could be approximated by aggregating data from accelerometers or EMG sensors on the conductor's arms and shoulders over a short-time sliding window. Alternately, the vertical position of the conductor's arm could be obtained from video, and this data characterized over time (integrated to derive velocity and acceleration, or otherwise aggregated) to model the conductor's physical effort during the performance. A previous study made some progress toward a method to determine the vertical position of the conductor's arm by applying computer vision hand-tracking techniques to a video recording of the concert featured in this study (Nakra, Salgian, & Pfirrmann, 2009; Salgian, Pfirrmann, & Nakra, 2007). Further work is needed to extrapolate effort from the position information that was obtained. (A demonstration video from that study is available at http://www.youtube.com/watch?v=KOVWlPv3DYs.)

The HR signal is a complex measure that reflects interactions between physical, emotional, and cognitive experiences. It is possible that the conductor's HR in this study reflects all three components: the physical effort of conducting, the emotional intensity response (attributed to the autonomic nervous system), and the cognitive load required to lead a large ensemble in front of a large audience. In order to investigate the various possible contributions to the conductor's HR variation, the authors animated the conductor's HR data and aligned it with a video recording that was taken of the performance of Mozart's *Marriage of Figaro* Overture. In this demonstration, the conductor's HR varies between a baseline level of 100 BPM to a peak level above 130 BPM. Features in the HR signal align with transitional features in the musical structure. This video can be accessed online at www.youtube.com/watch?v=ij2PIEiMOXw. More research is needed to distinguish and estimate the relative contributions of the different factors (effort, emotion, and cognitive load) in a conductor's HR signal.

It is also possible that a small amount of noise remained in the HR signal, post-processing. The raw HR signal itself contained a high amount of variability from second to second. Some of this variation was inherent to the computation approach that was used; instantaneous interpulse intervals determined the rate, which varied over time. However, some of the high frequency signal spikes and dips were attributed to noise and anomalies from low ECG pulse detection. Common causes for such anomalies could be experimental error (sensor leads temporarily detaching due to movement) or physiological causes, such as premature ventricular contraction.

The thematic and harmonic analysis that was performed on the musical score may also have some inherent limitations. While it revealed structural characteristics in the music that were reflected in the contour of the conductor's HR, the analysis was performed visually and post hoc. In addition, any musical score analysis might be said to be somewhat subjective and open to multiple interpretations. Finally, it remains unresolved whether the emotional intensity of the musical performance originated largely in the patterns and structures in the score, or from the interpretational choices made by the conductor in shaping those patterns during the concert. There nonetheless appears to be a temporal alignment between patterns in the score structure, the conductor's heart rate variations, and the audience response.

In the process of arriving at the conclusions in this paper, the authors also considered the potential concern that "continuous rating responses to music do not lend themselves to statistical significance tests of correlations, nor is this class of relatedness always the most relevant, particularly after compensating for serial correlation." (Upham, 2012) The problem identified by Upham is the statistical limitation of correlation analysis when used with non-independent data sets. Dean and Bailes (2010) addressed this issue by applying a new analytical tool set; time series analysis (TSA). More recently, they applied TSA to relate electroacoustic music patterns with listeners' real-time perceptual responses (Bailes & Dean, 2012), and to relate skin conductance and performer movement during piano improvisation (Dean & Bailes, 2013). TSA ensures stationarity by applying a simple differencing transform to original data sets, whereby subsequent data points in the time series are subtracted. The transformed data are then "pre-whitened," whereby an autoregressive model is fit to the data such that the residuals are free of any autocorrelation. The lag and coefficient parameters of the autoregressive model are used to model the other time series. Cross-correlation analysis is then performed on the residuals of both autoregressive models.

Although data transforms used with TSA may improve Type 1 error estimates, the modification of the original data may simultaneously increase Type II errors

(Pyper & Peterman, 1998). Additionally, these methods can alter the character of the original data, further complicating the interpretation of the results. For these reasons, and since the application of TSA to music perception studies is still somewhat new, the authors opted for a more conservative approach in the current study and applied the standard practice to use cross-correlations in comparing different sequences of continuous responses to music and characterizing their temporal relationship.

One important contribution of this study was to measure physiological phenomena and emotional responses in a real-world context. Live orchestral concerts provide an especially rich example of the "social context of music listening" (Juslin & Laukka, 2004) – one in which emotional responses can be intense. A central aspect of this approach was to ensure that the experiment design allowed participants to engage in authentic musical experiences that elicited genuine and optimal responses. While surface electrode sensors and other continuous measurement devices have been used for decades in emotion research, they are typically used in a laboratory setting. Some research indicates that emotions performed in a laboratory may be problematic (Hoque & Picard, 2011). In contrast to the lab environment, the social demands of a live performance situation may have motivated the participants to express genuine emotions. By integrating the ECG sensor into the conductor's clothing, the conductor could participate in the public concert without significant constraint, responding normally while conducting the music. In general, the participants expressed interest in the scientific experiment and its potential outcomes.

Compared with the number of studies of affect during music listening, there are very few examples of research on affect while performing and/or conducting music. Perhaps one reason for the lack of research in this area is the inherent difficulty in obtaining genuinely expressive affective data from moving participants without constraining their natural responses. For this study, researchers resolved several substantial logistical and technical challenges in order to gather data during a live professional concert. Whereas listener studies are typically administered to seated, stationary individuals, this study measured thirty-three simultaneous participants (including one very active conductor) in a large public venue. In this regard, the wearable ECG monitoring system used in this study was successful; the conductor was able to conduct normally throughout the concert. In this way, a laboratory-style experimental methodology was deployed into a real-world context. Artistic demands of the concert presentation were respected and observed, while sensitively integrating scientific measurement techniques. While it was challenging to balance the sometimes contradictory demands of experimental control with the complexity and uncontrollable variables of research in an ecologically valid setting, it was determined to be important to undertake this research in the realistic context provided by the live performance.

Why are emotional responses such important and integral parts of musical experiences for both performers and audiences? Perhaps it is because of the physiological responses to music that are accompanied by experiences commonly reported as "chills," "goosebumps," relaxation, or changes in energy level (Grewe, Nagel, Kopiez, & Altenmüller, 2007; Panksepp, 1995). These experiences are often associated with physiological effects such as changes in pulse rate, respiration rate and amplitude, blood pressure, electrodermal activity, and decreased muscular fatigue (Mursell, 1937, pp. 27-28). Physiological and affective experiences influence human behavior, including group identification and bonding (Meyer, 1956), muscle synchronization (Repp, 2005), exercise motivation and performance (Waterhouse, Hudson, & Edwards, 2010), pre-surgical tension reduction (Mok & Wong, 2003), and mental focusing and relaxation (Thompson, Schellenberg, & Husain, 2001). Nonetheless, the reasons why audiences and musicians jointly feel the push and pull of emotional intensity during concerts remains somewhat mysterious.

Conclusions

This study was undertaken in order to better understand whether conductors and audiences shared similar emotional responses during a musical performance. While most prior work has focused on listener responses to *recorded* music, this study sought to extend those results to a *realtime performance* setting, in a concert of the Boston Symphony Orchestra. Affective responses were compared across data from a conductor and thirty-two audience members. Using commonly accepted measures of emotional intensity, the data revealed statistically significant temporal correlations between the conductor's and audience's emotional intensities. The results indicate that: 1) variations in the conductor's HR were temporally aligned with structural patterns in the musical scores; and 2) these variations were highly correlated with the average of the audience's emotional intensity measurement. The highest degree of correlation between the conductor's HR and audience intensity measurement occurred during the first piece on the concert: $r(30) = .86$, $p < .001$. The first four

selections in the concert were studied; all four had significant positive correlations, the average of which was $r = .70$. Additionally, a qualitative analysis suggested features in the conductor and audience responses appeared to be temporally aligned with each other and with the musical score.

This study combined methods including self-report, physiological measure, and musical analysis. The conductor and audience participants were measured simultaneously. Implications for music perception include the interactions between audiences and musicians during concerts, differing emotional responses between performers and listeners, the impact of visual vs. auditory stimuli on listeners, and the relative impact of effort and affect in the heart rate of conductors.

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Correspondence concerning this article should be addressed to Teresa Marrin Nakra, Department of Music, The College of New Jersey, P.O. Box 7718, Ewing, NJ. E-mail: nakra@tcnj.edu

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